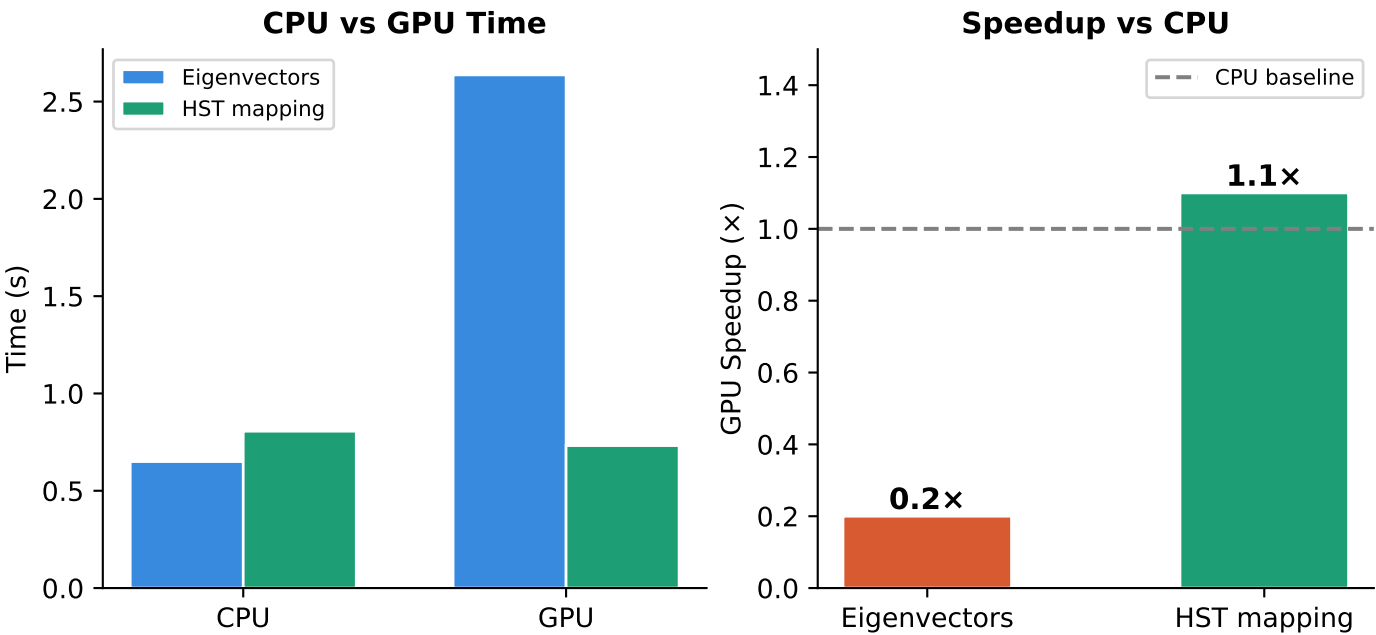


HST GPU Benchmark — Findings

NVIDIA RTX 4070 | FAUST dataset | 6890 vertices | May 2026

Benchmark Results

Component	CPU	GPU	Speedup
Eigenvectors (k=2)	0.650s	2.637s	0.2×
HST mapping	0.806s	0.732s	1.1×
Total	~0.8s	~3.4s	0.2×



Conclusion

CPU is faster than GPU for FAUST-scale meshes (6890 vertices).

Why GPU does not help

Eigenvectors: scipy ARPACK computes only k=2 sparse eigenvectors.
GPU eigh computes all 6890 — fundamentally slower for small k.
HST mapping: 6890 values too small to offset GPU transfer overhead.

When GPU would help

Meshes with 100k+ vertices
Batch processing of hundreds of pairs in parallel
ZoomOut with large k_final (k=200+)

Recommendation

Use CPU backend for FAUST-scale benchmarks.
GPU support available in add-on for future large-mesh testing.
Add-on auto-detects and falls back to CPU if GPU is slower.